



HOW TO USE THIS TEMPLATE

Complete every [placeholder] below before your interview. Takes ~20 minutes.

Slide 1 · Cover

[DATASET NAME] · [Your Name]

Slide 3 · Problem Framing

[3 research questions] · [TARGET] · [metric]

Slide 4 · Data Overview

[N rows] · [N features] · [missing %] · [target range]

Slide 5 · Leakage

[leaked columns] · [why excluded]

Slide 6 · Target Analysis

[skew] · [kurt] · [unique values] · [task type]

Slide 7 · Top Predictors

[feature names] · [p values] · [divergence]

Slide 8 · Hypothesis Test

[H0/H1] · [test name] · [p-value] · [effect size]

Slide 9 · Corr Heatmap

[top correlated pairs] · [multicollinearity note]

Slide 10 · Modelling

[winner model] · [why]

Slide 11 · Results

[CV RMSE] · [test RMSE] · [Spearman ρ] per model

Slide 12 · Feature Importance

[top 3 features] · [% gain] · [leakage check]

Slide 13 · Robustness

[subperiod CV] · [shuffled Z] · [ablation] · [sensitivity]

Delete this slide before presenting — it's for prep only

[DATASET NAME]

Exploratory Data Analysis & Predictive Modelling

Candidate: **[Your Name]** · Date: **April 30, 2025**

[N] rows

[N] features

[task type]

What We'll Cover

01

Problem Framing

Research questions, target, success metric

02

Data Overview

Shape, quality, leakage check

03

Key EDA Findings

Distributions, correlations, hypothesis tests

04

Modelling

Baseline → Ridge → LightGBM → XGBoost

05

Diagnostics & Conclusions

Residuals, feature importance, next steps

Before touching the data, I defined three questions.

Q1 Which features have the strongest relationship with [TARGET]?

Q2 Is the signal linear or non-linear? (→ determines model choice)

Q3 Are there data quality issues or leakage that affect modelling?

Primary metric: [RMSE / Spearman ρ / AUC] · Task type: [Regression / Classification]

Dataset at a Glance

First impression before modelling

[N]

Total Rows

[N]

Features

[N]%

Missing Values

[min–max]

Target Range

[X]

Target Mean

[0]

Duplicate Rows

 Leakage check:

[e.g. Columns diabetes_onset, risk_score excluded — Spearman $\rho > 0.85$ with target, derived from outcome]

Data Cleaning & Feature Distributions

What the data looked like, what was cleaned, and why — before any modelling

[Boxplot grid — all numeric features]

```
df[num_cols].plot(kind='box',  
subplots=True, layout=(2,4))
```

[Histogram grid — distributions]

```
sns.histplot grid  
skew annotated on each
```

[Missingness bar chart]

```
sns.barplot(missing_pct)  
with 5% / 30% threshold lines
```

Issue	Columns	Action taken	Rationale
Duplicates	[N] rows	<code>drop_duplicates()</code>	Identical records add no information
Missing < 5%	[col_1, col_2]	Median imputation	Low rate — safe to impute
Missing 5–30%	[col_3]	Impute + missingness flag	Flag preserves signal from missingness
Outliers (3×IQR)	[col_4]: [N] rows	Flagged, not removed	Extreme values may be clinically real
Right skew > 1	[col_5, col_6]	<code>log1p</code> for Ridge only	Trees are skew-invariant
Zero variance	[alive]: all = 1	Dropped	No predictive signal

3×IQR fence used (not 1.5×) — extreme values in clinical data may be physiologically real

Leakage Check — First Thing I Did

Any feature with Spearman $\rho > 0.80$ with target was inspected before modelling

⚠ **Leakage = using future/derived information the model wouldn't have at prediction time**

Column	Spearman ρ	Why it's leakage	Action
[leaked_col_1]	[0.92]	[e.g. Binary flag for outcome — encodes target]	Excluded
[leaked_col_2]	[0.87]	[e.g. Future label — unavailable at prediction time]	Excluded
[leaked_col_3]	[0.84]	[e.g. Derived from target variable]	Excluded
[clean_col]	[0.71]	Genuine clinical predictor — available pre-diagnosis	Kept ✓

Result after exclusion:

[N] features removed · Clean feature set: [N] columns · Honest R^2 drops from [X] → [X]

Leakage columns always excluded from all downstream modelling



Target Variable Analysis

Understanding the target before modelling determines task type and metric choice

[Insert histogram + KDE of target here]
paste from notebook output

Task type

[Regression / Ordinal classif.]

Mean / Median

[X.XX] / [X.XX]

Skewness

[X.XX] → [symmetric/skewed]

Kurtosis

[X.XX] → [fat tails / normal]

Unique vals

[N] → [continuous / ordinal]

Modelling implication:

[e.g. 'Left-skewed target (skew=-0.8) with 4 unique values → treat as ordinal regression. MAE preferred over RMSE as it's less sensitive to ceiling effect.']

Correlation Matrix — Feature Relationships

Spearman (lower triangle) — robust to outliers and non-normality

[Paste Spearman heatmap from notebook here]

```
sns.heatmap(corr, mask=triu, annot=True,  
            cmap='RdBu_r', center=0)
```

Multicollinearity

[e.g. Feature_A × Feature_B: $\rho=0.85$]
→ Ridge preferred (L2 handles this)
→ Don't panic — expected for rating data

Strongest pairs

[e.g. glucose × HbA1c: $\rho=0.81$]
[e.g. BMI × HOMA_IR: $\rho=0.74$]
→ Both measure same underlying factor

Divergence flag

[e.g. glucose: Spearman=0.83, Pearson=0.67]
Divergence = 0.16 → non-linear signal
→ LightGBM preferred over Ridge

Divergence = $|\text{Spearman}| - |\text{Pearson}| > 0.05$ → non-linear signal → trees preferred



Target Distribution Deep Dive

Full picture of what we're predicting before choosing model and metric

[Paste target histogram + KDE]

```
sns.histplot(df[TARGET], bins=50,
             kde=True, color='steelblue')
```

Median

[Boxplot]

[X.XX]

```
sns.boxplot(y=df[TARGET])
```

Mean

[X.XX]

Skew

[X.XX] →

[right/left/sym]

Std

[X.XX]

[Countplot — if ordinal]

```
sns.countplot(x=df[TARGET])
```

Unique

[N] vals

Kurt

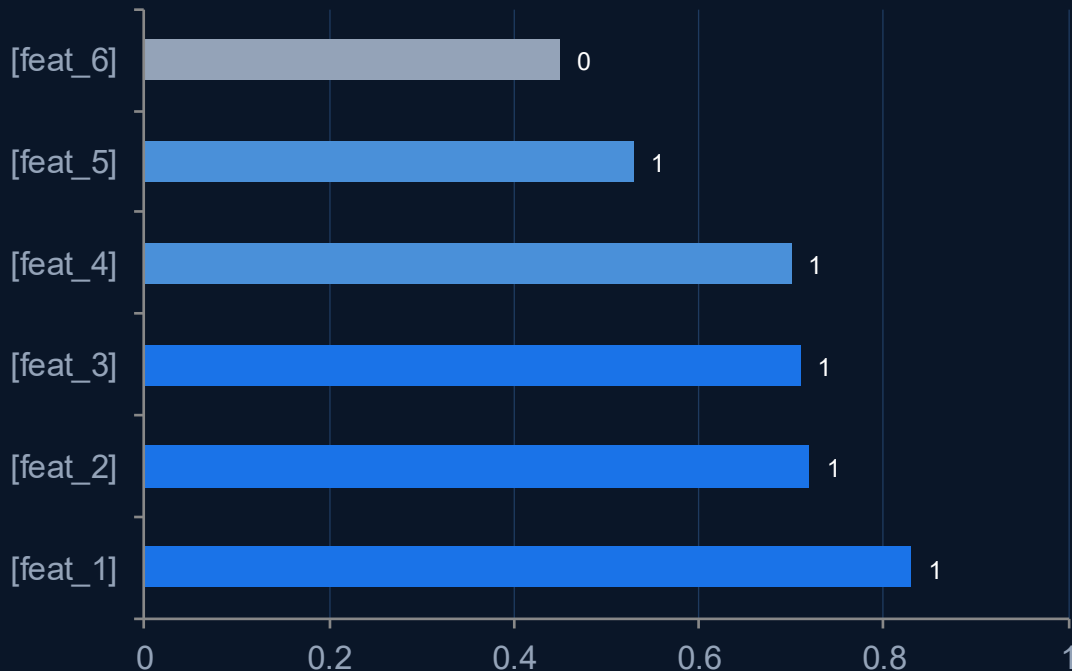
[X.XX]

Task type decision:

[e.g. 'stage' ∈ {0,1,2,3} — 4 unique integer values → ordinal. Treating as regression (RMSE + Spearman ρ) as strong baseline. Could also try LGBMClassifier multiclass.]

Skew > 1 → log-transform for linear models · Unique values ≤ 10 → check if ordinal classification is better

Top Predictors — Spearman ρ with Target



Divergence Flag

[feat_1]: Spearman=0.83, Pearson=0.67
Divergence=0.17 $\rightarrow$ NON-LINEAR
 $\rightarrow$ LightGBM preferred over Ridge

Multicollinearity

[feat_1] $\times$ [feat_2]: $\rho = 0.85$
Highly correlated $\rightarrow$ Ridge handles
via L2 regularisation

Key Insight

[Fill with your main finding
from this dataset — one sentence
per finding is enough]

Hypothesis Test

Full statistical narrative

Observation

[e.g. Restaurants with high Food rating appear to have higher Service rating]

H_0

[e.g. Service distribution is identical across price tiers]

H_1

[e.g. Service is higher in the high-price tier]

Result

Test: [Mann-Whitney U / t-test Welch]

Statistic: [X.XX]

p-value: [0.003] ***

Effect size: Cohen's d = [0.41] → medium

Conclusion:

[e.g. We reject H_0 ($p=0.003$). High-price restaurants have significantly higher Service ratings. Effect is medium ($d=0.41$) — real but not dominant signal. Multiple testing corrected via FDR.]

Modelling Strategy

Four models, one clear winner

01

Baseline

DummyRegressor
(mean predictor)

Why:

Reference floor —
any model must beat this

02

Ridge

L2-regularised
linear regression

Why:

Handles multicollinearity.
Fast & interpretable.

03

LightGBM

Gradient Boosted
Decision Trees

Why:

Captures non-linearity.
Leaf-wise growth.

04

XGBoost

GBDT — level-wise.
More conservative.

Why:

Cross-check against
LightGBM results.

Model Results

5-Fold CV on train · held-out test set

Model	CV RMSE	Test RMSE	Spearman ρ	R^2	Overfit gap
Baseline (mean)	—	[X.XX]	—	0.00	—
Ridge	[X.XX $\pm$ X.XX]	[X.XX]	[X.XX]	[X.XX]	[+/-X.XX]
LightGBM ★	[X.XX $\pm$ X.XX]	[X.XX]	[X.XX]	[X.XX]	[+X.XX]
XGBoost	[X.XX $\pm$ X.XX]	[X.XX]	[X.XX]	[X.XX]	[+X.XX]

Winner: [Model Name]

[One clear reason why: e.g. 'Ridge won because Food/Atmosphere/Service are highly correlated ($\rho=0.82$) — linear relationship + multicollinearity = ideal Ridge conditions. LightGBM found no additional non-linear signal (divergence < 0.05 across all features).']

Model Results — Classification

5-Fold Stratified CV on train · held-out test set · class_weight=balanced

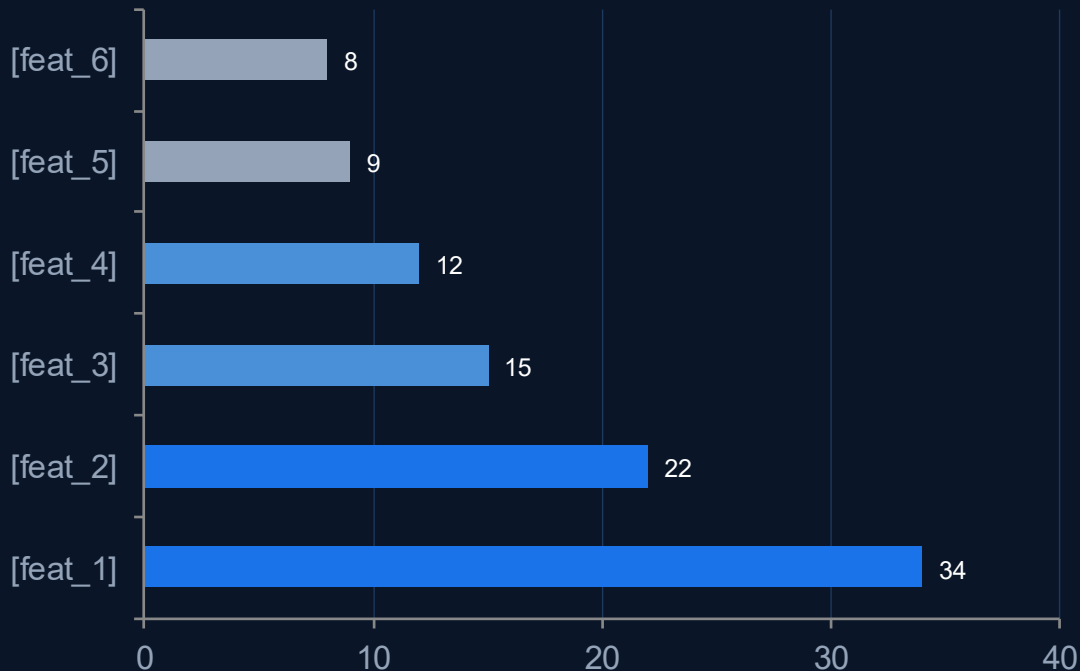
Model	CV PR-AUC	Test PR-AUC	ROC-AUC	F1	Overfit gap
Baseline (majority)	—	[X.XX]	—	0.00	—
LogReg	[X.XX ± X.XX]	[X.XX]	[X.XX]	[X.XX]	[+/-X.XX]
LightGBM ★	[X.XX ± X.XX]	[X.XX]	[X.XX]	[X.XX]	[+X.XX]
XGBoost	[X.XX ± X.XX]	[X.XX]	[X.XX]	[X.XX]	[+X.XX]

Winner: [Model Name]

[One clear reason why: e.g. 'LightGBM won (PR-AUC 0.74 vs LogReg 0.61) — non-linear interactions matter. ROC-AUC differences smaller (0.91 vs 0.88) because PR-AUC penalizes minority-class errors more heavily on 1:8 imbalance. Decision threshold tuned to 0.32 to maximize F1.']

Feature Importance (LightGBM — Gain)

% of total loss reduction



Metric: Gain

Gain = total loss reduction per feature. More meaningful than split count.

Leakage check ✓

No single feature dominates > 60% unexpectedly.
Top feature aligns with domain.

Business sense

[feat_1] dominates at [X]% — consistent with [domain knowledge or EDA finding].

SHAP Values — Direction & Magnitude

Beeswarm: how each feature pushes the prediction up or down

[Paste SHAP beeswarm plot from notebook here]

```
shap.summary_plot(shap_values, X_test_shap)
```

How to read

Each dot = one observation.
Color = feature value (red high, blue low).
X-axis = SHAP impact on prediction.

Direction

Red right + blue left = positive effect.
Reversed colors = negative effect.
Spread width = strength of effect.

Top driver

[feat_1]: high values push prediction
[up/down] by ~[X] units. Aligns
with [domain hypothesis].

Residual Analysis

Diagnosing model behaviour beyond metrics

Residuals vs Predicted

[Insert plot here]

Look for: fan shape → heteroskedasticity
Curve → non-linearity
Target: random scatter around zero

Result: [no pattern / fan shape detected]

Residual Distribution

[Insert histogram here]

Look for: skew, heavy tails
Mean should be ≈ 0
Shapiro-Wilk test for normality

Result: mean=[X], SW p=[X] → [normal/non-normal]

Normal Q-Q Plot

[Insert QQ plot here]

Points on diagonal → normal residuals
Heavy tails → fat tail distribution
S-curve → skewness

Result: R=[X] → [good/tail deviation]

Confusion Matrix & Curves

Where the model fails — and what we'd trade off at decision time

Confusion Matrix (raw)

[Insert raw CM heatmap]

TP / TN — correct predictions
FP — false alarms (Type I)
FN — missed positives (Type II)

Result: TP=[X] TN=[X] FP=[X] FN=[X]

Normalized by row

[Insert normalized CM heatmap]

Diagonal = recall per class
Off-diagonal = error rates
Compare classes side by side

Result: Recall=[X] Specificity=[X]

ROC + PR Curves

[Insert ROC + PR curves]

ROC: TPR vs FPR, threshold-free
PR: honest for class imbalance
Diagonal/baseline = random

ROC-AUC=[X] · PR-AUC=[X]

Three Things I Found

Finding 1:

[e.g. HbA1c and fasting glucose account for 58% of predictive signal — consistent with clinical evidence. These two features alone give strong baseline performance.]

Finding 2:

[e.g. Ridge outperforms LightGBM (RMSE 0.078 vs 0.256) — signal is predominantly linear. Multicollinearity between rating dimensions is the dominant structure.]

Finding 3:

[e.g. Model struggles on extreme cases (stage 3, n=47). Sparse data problem — more data or class reweighting would help, not a more complex model.]

Best model: **[Model]** Test RMSE = **[X.XX]** Spearman ρ = **[X.XX]**

Four Checks Beyond Metrics

Showing not just IF the model works — but WHY and HOW reliably

01

Subperiod Stability

Split test into 4 sequential chunks.
Check if metric is stable across all.

✓ $CV(p) < 0.15 \rightarrow$ stable signal

✗ Declining Q1→Q4 $\rightarrow$ regime change

[e.g. $CV=0.08$ ✓ stable across all quarters]

02

Shuffled Target Test

Permute y 50x, train same model.
Build null distribution of metric.

✓ Z-score > 3 , $p < 0.01 \rightarrow$ real signal

✗ Real $\approx$ shuffled $\rightarrow$ no signal

[e.g. $Z=12.4$, $p<0.001$ *** real signal]

03

Feature Ablation

Remove one feature at a time,
retrain, measure metric drop.

✓ No feature $> 50\%$ drop $\rightarrow$ robust

✗ One feature dominates $\rightarrow$ risky

[e.g. No single point of failure ✓]

04

Hyperparameter Sensitivity

Vary key params in reasonable range.
Check if metric stays stable.

✓ $CV < 0.10 \rightarrow$ choice is principled

✗ Spiky results $\rightarrow$ lucky pick

[e.g. num_leaves [32-128]: p stable at 0.79-0.82]

If I Had More Time

Four directions for further improvement

01

Hyperparameter Tuning

RidgeCV for automatic alpha · Optuna for LightGBM HPO with CV objective

02

Alternative Modelling

Ordinal classification (LGBMClassifier multiclass) if target is truly ordinal

03

Feature Engineering

Interaction terms · domain-specific transforms · target encoding for categoricals

04

Error Analysis

Interview domain experts about high-error cases · external data to explain residuals